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SUBSTRATE TRANSFER SYSTEM

Abstract

The provided is a substrate transfer system that can prevent the mirror-like main surface of the substrate from coming into contact with the system components during transfer and causing scratches on the main surface. The substrate transfer system has a system component; a transfer section that moves a substrate relative to the system component; an imaging section that captures a mirror image of the system component reflected on a mirror-like main surface of the substrate and a real image of the system component, when the substrate is moved by the transfer section; and a judgment section that performs an abnormality judgment to determine whether any abnormality exists in clearance between the main surface of the substrate and the system component, based on the mirror image and the real image captured by the imaging section.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to a substrate transfer system.

BACKGROUND

[0002] A substrate transfer system that moves substrates is known (see, for example, Patent Document 1).

CITATION LIST

Patent Literature

[0003] PTL 1: JP 2013-110200 A1

SUMMARY

[0004] In the above-mentioned substrate transfer system, it is desirable to be able to prevent the mirror-like main surface of the substrate from coming into contact with a system component during transfer and causing scratches. To achieve this, it is thought that an imaging section could be installed to capture the substrate and the system component from directly beside the substrate, and monitor the clearance between the main surface of the substrate and the system component based on the resulting images. However, in this case, it was difficult to stably monitor the clearance due to the thin substrate and the effects of reflected light, etc.

[0005] The object of the present disclosure is to provide a substrate transfer system that can prevent the mirror-like main surface of the substrate from coming into contact with the system components during transfer and causing scratches on the main surface.

[0006] One aspect of the present disclosure is as follows. [0007] [1] A substrate transfer system, having [0008] a system component; [0009] a transfer section that moves a substrate relative to the system component; [0010] an imaging section that captures a mirror image of the system component reflected on a mirror-like main surface of the substrate and a real image of the system component, when the substrate is moved by the transfer section; and [0011] a judgment section that performs an abnormality judgment to determine whether any abnormality exists in clearance between the main surface of the substrate and the system component, based on the mirror image and the real image captured by the imaging section. [0012] [2] The substrate transfer system as described in [1], wherein the judgment section performs the abnormality judgment based on apparent clearance between the mirror image and the real image in an image captured by the imaging section. [0013] [3] The substrate transfer system as described in claim [1] or [2], wherein the judgment section performs the abnormality judgment based on a change between: the clearance calculated for the substrate as a preceding substrate that approaches the system component by the transfer section, and the clearance calculated for the substrate as a succeeding substrate that approaches the system component by the transfer section after the preceding substrate has retreated from the system component. [0014] [4] The substrate transfer system as described in any one of [1] to [3], having a notification section that performs notifications when an abnormality is detected through the abnormality judgement. [0015] [5] The substrate transfer system as described in claim 4, having [0016] a second system component; [0017] a second transfer section that moves a second substrate relative to the second system component; and [0018] a second imaging section that captures a mirror image of the second system component reflected on a mirror-like main surface of the second substrate and a real image of the second system component, when the second substrate is moved by the second transfer section; wherein [0019] the judgement section performs a second abnormality judgment to determine whether any abnormality exists in clearance between the main surface of the second substrate and the second system component, based on the mirror image and

the real image captured by the second imaging section, and the notification section performs notifications when an abnormality is detected through the second abnormality judgment.

[0020] According to the present disclosure, it is possible to provide a substrate transfer system that can prevent the mirror-like main surface of the substrate from coming into contact with the system components during transfer and causing scratches on the main surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] In the accompanying drawings:

[0022] FIG. 1 is a top view of a first transport processing unit of the substrate transfer system according to one embodiment of the present disclosure;

[0023] FIG. 2 is a cross-sectional view along the line A-A when a first substrate is moved onto a first transport pallet by a first transfer section;

[0024] FIG. 3 is a schematic diagram illustrating an image captured by a first imaging section illustrated in FIG. 1;

[0025] FIG. 4 is a schematic diagram illustrating the overall substrate transfer system; and

[0026] FIG. 5 is a graph providing an example of the relationship between the apparent clearance and the actual clearance in the image illustrated in FIG. 3.

DETAILED DESCRIPTION

[0027] Hereinafter, an embodiment of the present disclosure will be described with reference to the drawings.

[0028] As illustrated in FIGS. 1 to 4, in one embodiment of the present disclosure, a substrate transfer system 1 (see FIG. 4) has a (first) system component 6A (see FIG. 1) that forms part of the substrate transfer system 1; a (first) transfer section 3A that moves a circular (first) substrate 11A relative to the system component 6A; a (first) imaging section 7A (see FIG. 1) that captures a mirror image 12 of the system component 6A reflected on a mirror-like main surface of the substrate 11A and a real image 13 of the system component 6A, when the substrate 11A is moved by the transfer section 3A (see FIG. 2) relative to the system component 6A; and a judgment section 8a (see FIG. 4) that performs a (first) abnormality judgment to determine whether any abnormality exists in clearance C (see FIG. 2) between the main surface of the substrate 11A and the system component 6A, based on the mirror image 12 and the real image 13 (see FIG. 3) captured by the imaging section 7A.

[0029] As illustrated in FIG. 4, in this embodiment, the substrate transfer system 1 has: a (first) transport processing unit 2A, which has the (first) transfer section 3A, a (first) transport pallet 4A, and a (first) processing section 5A; and a (second) transport processing unit 2B, which has a (second) transfer section 3B, a (second) transport pallet 4B, and a (second) processing section 5B. The (first) transport pallet 4A has a (first) gripper 4aA as the (first) system component 6A and a (first) pallet 4bA. The second transport pallet 4B has a second gripper 4aB as a second system component 6B and a second pallet 4bB. In addition, the substrate transfer system 1 has the third transport processing unit 2C, the fourth transport processing unit 2D, the fifth transport processing unit 2E, and the sixth transport processing unit 2F. The number of transport processing units in the substrate transfer system 1 can be set as appropriate. The (first) transport processing unit 2A transports and processes the (first) substrate 11A, the (second) transport processing unit 2B transports and processes the (second) substrate 11B, the (third) transport processing unit 2C transports and processes the (third) substrate 11C, the (fourth) transport processing unit 2D transports and processes the (fourth) substrate 11D, the (fifth) transport processing unit 2E transports and processes the (fifth) substrate 11E, and the (sixth) transport processing unit 2F transports and processes the (sixth) substrate 11F.

[0030] The substrate transfer system **1** has the (first) imaging section **7A** for the (first) transport processing unit **2A**, a (second) imaging section **7B** for the second transport processing unit **2B**, the judgment section **8A**, and a notification section **9**. The judgment section **8a** is capable of communicating with the notification section **9**, the (first) imaging section **7A**, and the second imaging section **7B**. In addition, the substrate transfer system **1** has a third imaging section **7C** for the third transport processing unit **2C**, a fourth imaging section **7D** for the fourth transport processing unit **2D**, a fifth imaging section **7E** for the fifth transport processing unit **2E**, and a sixth imaging section **7F** for the sixth transport processing unit **2F**.

[0031] As illustrated in FIG. 2, when the transfer section **3A** moves the substrate **11A** (such as a silicon wafer) against the system component **6A**, if there is no vertical clearance **C** between the main surface of the substrate **11A** (in this embodiment, the top surface of the circular substrate **11A**) and the system component **6A**, the main surface of the substrate **11A** will come into contact with the system component **6A**, and scratches will appear on the main surface of the substrate **11A**. Conversely, if the clearance **C** is too large, the main surface of the other side (under surface) of the substrate **11A** may come into contact with other system components, and scratches may appear on the main surface of the other side of the substrate **11A**. For this reason, the transfer section **3A** is controlled by a control device (not illustrated) composed of a computer or other components so that the appropriate clearance **C**, which is adjusted in advance, can be allowed during the transfer of the substrate **11A**. However, there are cases where the clearance **C** decreases due to abnormalities in the (first) gripper **4A** or due to a malfunction of the robot arm that composes the (first) transfer section **3A** or the control device that controls the robot arm. In this embodiment, rather than using the imaging section **7A** to capture the substrate **11A** and the system component **6A** from directly beside the substrate **11A**, the imaging section **7A** captures them from an oblique angle above the substrate **11A**. This allows, as illustrated in FIG. 3, the mirror image **12** of the system component **6A** reflected on the mirror-like main surface to be captured in a single image **14** together with its real image **13**. Thus, the clearance **C** can be stably monitored based on the positional relationship between the mirror image **12** and the real image **13** in the image **14**. Therefore, it is possible to realize a substrate transfer system **1** that can prevent the mirror-like main surface of the substrate **11A** from coming into contact with the system component **6A** during transfer and causing scratches on the main surface.

[0032] The judgment section **8a** performs the abnormality judgment based on apparent clearance **Ca** between the mirror image **12** and the real image **13** in the image **14** captured by the imaging section **7A**. According to the above configuration, the clearance **C** can be monitored more stably based on the apparent clearance **Ca** (the distance between the mirror image **12** and the real image **13**) in the image **14**. The apparent clearance **Ca** correlates with the actual clearance **C**, as illustrated in FIG. 5, so it is possible to monitor the clearance **C** based on the apparent clearance **Ca**. FIG. 5 is a graph providing an example of the relationship between the apparent clearance **Ca** detected by image processing and the actual measured clearance **C**. The abnormality judgment is performed, for example, by comparing the calculated value of the apparent clearance **Ca** with a threshold value. The judgment section **8a** is composed of a processing device **8** (computer, etc.) that can perform image processing to detect the apparent clearance **Ca**.

[0033] The imaging section **7A** is composed of a camera, etc., such as a CCD (Charge Coupled Device) image sensor. The image **14** captured by the imaging section **7A** may be a moving image or a still image.

[0034] The judgment section **8a** may be configured to perform the abnormality judgment based on a change between the clearance **C** calculated for the substrate **11A** as a preceding substrate that approaches the system component **6A** by the transfer section **3A** and the clearance **C** calculated for the substrate **11A** as a succeeding substrate that approaches the system component **6A** by the transfer section **3A** after the preceding substrate has retreated from the system component **6A**. According to the above configuration, it is possible to monitor the changes in the clearance **C**

between the preceding substrate and the succeeding substrate, therefore, to prevent the substrate **11A**, which approaches the system component **6A** by the transfer part **3A** after the succeeding substrate has retreated from the system component **6A** from coming into contact with the system component **6A** and causing scratches. The abnormality judgment is performed, for example, by comparing the change in the clearance **C** with a threshold value. In addition, the abnormality judgment may be performed based not only on the change in clearance **C**, but also on the clearance **C** calculated for the preceding or succeeding substrate.

[0035] The substrate transfer system **1** has the gripper **4aA** as the system component **6A** and the pallet **4bA**. The gripper **4aA** holds the substrate **11A** between a receiving part **4dA** of the pallet **4bA** and the gripper **4aA** itself by moving forward with respect to the pallet **4bA**, and releases the substrate **11A** by moving backward with respect to the pallet **4bA**. In addition to the gripper **4aA** and the pallet **4bA**, the transport pallet **4A** has a plurality of other grippers **4cA**. The number of other grippers **4cA** provided on the transport pallet **4A** is two in this embodiment, but this is not limited to this number. The transport pallet **4A** may be configured without the other grippers **4cA**. The gripper **4aA** and the other grippers **4cA**, respectively, can be made of a fluoro-resin, such as polytetrafluoroethylene, and others as a material. The receiving part **4dA** can be made of engineering plastics, such as Poly Ether Ether Ketone (PEEK), and others as a material.

[0036] The transfer section **3A** performs two types of transport: the first transport (see the white arrow pointing to the right in FIG. **1**) transports the substrate **11A** in the forward direction towards the system component **6A**, and the second transport (see the bold arrow pointing to the left in FIG. **1**) transports the substrate **11A** in the backward direction away from the system component **6A**.

[0037] The transfer section **3A** is composed of a robot arm that has a placement part **3aA** on which the substrate **11A** is placed. The substrate **11A** that has been placed on the placement part **3A** is placed on the pallet **4bA** of the transport pallet **4A** through the first transport performed by the transfer section **3A**, and is held between the gripper **4aA** and the pallet **4bA**. The substrate **11A** is transported into the processing section **5A** by the transport pallet **4A** in this state, and after the prescribed processing, it is returned to its original position by the transport pallet **4A** (see the double-headed bold arrow in FIG. **1**). Furthermore, the substrate **11A** is released due to the retraction of the gripper **4aA** and is transported from the transport pallet **4A** through the second transport performed by the transfer part **3A**.

[0038] The imaging section **7A** may be configured to constantly perform capturing to monitor the clearance **C**, or it may be configured to perform capturing only when necessary. In this case, for example, a sensor **15** that detects the first transport can be provided, and the processing device **8** that controls the imaging device **7A** can be linked to the sensor **15**, so that the imaging device **7A** can be configured to perform capturing only at the timing when the first transport is performed.

[0039] As illustrated in FIG. **4**, the substrate transfer system **1** has a notification section **9** that performs notifications when an abnormality is detected through the abnormality judgement. According to the above configuration, by notifying the system administrator **10** of the abnormality through the notifications, it is possible to alert them and let them take measures such as inspecting the substrate transfer system **1** or stopping its operation. The notification section **9** can be composed of a monitor **9a** that provides notification by displaying information on the screen, a warning light **9b** that provides notification by light, or a buzzer (not illustrated) that provides notification by sound, but this is not limited to these.

[0040] As illustrated in FIG. **4**, the substrate transfer system **1** has a second system component **6B** that forms part of the substrate transfer system **1**; a second transfer section **3B** that moves a second substrate **11B** relative to the second system component **6B**; and a second imaging section **7B** that captures a mirror image **12** of the second system component **6B** reflected on a mirror-like main surface of the second substrate and a real image **13** of the second system component **6B**, when the second substrate is moved by the second transfer section **3B**. The judgement section **8a** performs a second abnormality judgment to determine whether any abnormality exists in clearance **C** between

the main surface of the second substrate and the second system component **6B**, based on the mirror image **12** and the real image **13** captured by the second imaging section **7B**. The notification section **9** performs notifications when an abnormality is detected through the second abnormality judgment. According to the above configuration, the first abnormal judgment and its notification and the second abnormal judgment and its notification can be performed using the common judgment section **8a** and the notification section **9**, therefore, efficient operation of the entire system can be achieved. However, the substrate transfer system **1** may be configured not only as described above, using a common judgment section **8a**, but also, for example, with a separate first judgment section (not illustrated) for performing the first abnormality judgment and a separate second judgment section (not illustrated) for performing the second abnormality judgment. [0041] Although the embodiment of the present disclosure has been described above, the present disclosure is not limited to the above-described embodiment, and the above-described embodiment can be modified in various ways without departing from the gist of the present disclosure.

Claims

1. A substrate transfer system, having a system component; a transfer section that moves a substrate relative to the system component; an imaging section that captures a mirror image of the system component reflected on a mirror-like main surface of the substrate and a real image of the system component, when the substrate is moved by the transfer section; and a judgment section that performs an abnormality judgment to determine whether any abnormality exists in clearance between the main surface of the substrate and the system component, based on the mirror image and the real image captured by the imaging section.
 2. The substrate transfer system as described in claim 1, wherein the judgment section performs the abnormality judgment based on apparent clearance between the mirror image and the real image in an image captured by the imaging section.
 3. The substrate transfer system as described in claim 1, wherein the judgment section performs the abnormality judgment based on a change between: the clearance calculated for the substrate as a preceding substrate that approaches the system component by the transfer section, and the clearance calculated for the substrate as a succeeding substrate that approaches the system component by the transfer section after the preceding substrate has retreated from the system component.
 4. The substrate transfer system as described in claim 1, having a notification section that performs notifications when an abnormality is detected through the abnormality judgement.
 5. The substrate transfer system as described in claim 4, having a second system component; a second transfer section that moves a second substrate relative to the second system component; and a second imaging section that captures a mirror image of the second system component reflected on a mirror-like main surface of the second substrate and a real image of the second system component, when the second substrate is moved by the second transfer section; wherein the judgement section performs a second abnormality judgment to determine whether any abnormality exists in clearance between the main surface of the second substrate and the second system component, based on the mirror image and the real image captured by the second imaging section, and the notification section performs notifications when an abnormality is detected through the second abnormality judgment.
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